

A COUNTEREXAMPLE TO A REAL- p MIXED MATRIX NORM EXTENSION

FA_BANACH.001 COUNTEREXAMPLE PACKET

STATUS

This packet records a counterexample to a question in Sah–Sawhney–Stoner–Zhao, *A reverse Sidorenko inequality*, arXiv:1809.09462. In Section 3.1, after Lemma 3.1, they define

$$\|M\|_{L_{p,q}} = \left(\sum_i \left(\sum_j |m_{ij}|^p \right)^{q/p} \right)^{1/q}$$

and ask whether, for nonnegative matrices A, B , the inequality

$$(1) \quad \left\| A^\top B \right\|_{L_{p,q}}^2 \leq \left\| A^\top A \right\|_{L_{q,q}} \left\| B^\top B \right\|_{L_{p,p}}$$

extends from positive integer p to all real $1 \leq p \leq q$.

Here is the above proof written out in the language of matrices and vectors, with $\mathbf{1}_m$ denoting the all ones vector:

$$\begin{aligned} \|A^\top B\|_{L_{1,q}}^2 &= \|A^\top B \mathbf{1}_m\|_q^2 = \sup_{\|\mathbf{u}\|_{q'} \leq 1} \langle \mathbf{u}, A^\top B \mathbf{1}_m \rangle^2 = \sup_{\|\mathbf{u}\|_{q'} \leq 1} \langle A\mathbf{u}, B \mathbf{1}_m \rangle^2 \\ &\leq \sup_{\|\mathbf{u}\|_{q'} \leq 1} \langle A\mathbf{u}, A\mathbf{u} \rangle \langle B \mathbf{1}_m, B \mathbf{1}_m \rangle = \sup_{\|\mathbf{u}\|_{q'} \leq 1} \langle \mathbf{u} \mathbf{u}^\top, A^\top A \rangle \langle \mathbf{1}_m \mathbf{1}_m^\top, B^\top B \rangle \leq \|A^\top A\|_{L_{q,q}} \|B^\top B\|_{L_{1,1}}. \end{aligned}$$

We do not know if the inequality can be extended to $\|A^\top B\|_{L_{p,q}}^2 \leq \|A^\top A\|_{L_{q,q}} \|B^\top B\|_{L_{p,p}}$ for all reals $1 \leq p \leq q$ (this is true for positive integer p by a tensor-power argument). Note that the above proof holds for all real matrices A and B provided that $A^\top A, B^\top B, A^\top B$ have nonnegative entries.

IDEA OF THE PROOF

When $p = q$, the mixed norm is just the entrywise ℓ_p -norm. For integer p , the kernel $\langle x, y \rangle^p$ is positive semidefinite because it is the inner product of tensor powers. The natural place to look for a non-integer obstruction is therefore a small positive-measure configuration for the kernel $\langle x, y \rangle^p$ on the positive cone. Two directions in $\mathbb{R}_+^2$, with reciprocal weights, already break the Cauchy–Schwarz-type inequality at $p = 3/2$.

COUNTEREXAMPLE

Set

$$p = q = \frac{3}{2}, \quad t = \frac{1}{4}, \quad r = 1 - t^p = \frac{7}{8}, \quad \alpha = r^{1/p} = r^{2/3}.$$

Let

$$A = \begin{pmatrix} 0 & 1 \\ \alpha & t \end{pmatrix}, \quad B = \begin{pmatrix} t & \alpha \\ 1 & 0 \end{pmatrix}.$$

These are nonnegative real 2×2 matrices.

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PROOF

Since $p = q$, write $\|M\|_p = (\sum_{ij} |m_{ij}|^p)^{1/p}$. The proposed inequality (1) becomes

$$\left\|A^\top B\right\|_p^2 \leq \left\|A^\top A\right\|_p \left\|B^\top B\right\|_p.$$

The columns of A are αe_2 and $(1, t)$, while the columns of B are $(t, 1)$ and αe_1 . Thus

$$\left\|A^\top B\right\|_p^p = (\alpha)^p + (2t)^p + (\alpha)^p = 2r + (2t)^p =: S.$$

Similarly,

$$\left\|A^\top A\right\|_p^p = \alpha^{2p} + 2(\alpha t)^p + (1 + t^2)^p = r^2 + 2rt^p + (1 + t^2)^p =: X.$$

By symmetry $\left\|B^\top B\right\|_p^p = X$. Therefore (1) is equivalent to

$$S^{2/p} \leq X^{2/p},$$

or simply $S \leq X$.

For $t = 1/4$ and $p = 3/2$,

$$S = 2 \cdot \frac{7}{8} + \left(\frac{1}{2}\right)^{3/2} = \frac{7 + \sqrt{2}}{4},$$

whereas

$$X = \left(\frac{7}{8}\right)^2 + 2 \cdot \frac{7}{8} \cdot \frac{1}{8} + \left(\frac{17}{16}\right)^{3/2} = \frac{63 + 17\sqrt{17}}{64}.$$

It remains only to compare these two numbers. Multiplying by 64, the claim $S > X$ is

$$112 + 16\sqrt{2} > 63 + 17\sqrt{17},$$

or

$$49 + 16\sqrt{2} > 17\sqrt{17}.$$

This holds because

$$(49 + 16\sqrt{2})^2 - (17\sqrt{17})^2 = 1568\sqrt{2} - 2000 > 0,$$

and the final inequality follows from $\sqrt{2} > 125/98$. Hence $S > X$, so

$$\left\|A^\top B\right\|_{L_{3/2,3/2}}^2 > \left\|A^\top A\right\|_{L_{3/2,3/2}} \left\|B^\top B\right\|_{L_{3/2,3/2}}.$$

This disproves the proposed real- p extension.

VERIFICATION NOTES

The script `code/verify_counterexample.py` evaluates the same example numerically and prints the ratio

$$\frac{\left\|A^\top B\right\|_{L_{3/2,3/2}}^2}{\left\|A^\top A\right\|_{L_{3/2,3/2}} \left\|B^\top B\right\|_{L_{3/2,3/2}}},$$

which is approximately $1.0154 > 1$. The proof above is exact; the script is only a check against transcription errors.

NOVELTY AND LIMITATIONS

Cheap run indexes were searched for arXiv:1809.09462, “reverse Sidorenko”, “mixed norm”, and the exact matrix inequality; no prior run packet or attempt was found. Local parsed-source search found the exact inequality only in the source paper and generated target-pool extracts. Web searches on 2026-07-03 for exact phrases from the inequality and the source sentence found no visible later answer. This is a bounded novelty check, not a publication claim.

The counterexample only disproves the proposed extension to all real p . It does not contradict the source paper’s $p = 1$ lemma, the positive-integer cases obtained by tensor powers, or any of the reverse Sidorenko theorems.

REFERENCE

A. Sah, M. Sawhney, D. Stoner, and Y. Zhao, *A reverse Sidorenko inequality*, arXiv:1809.09462.